

Sensorless Vector Control of Permanent Magnet Motors Using Observer-Based Adaptive Estimation Techniques

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ABSTRACT

Occurring widely in modern highly-performance drive systems like robotics, electric vehicles, aerospace mechanisms and high precision manufacturing equipment, Permanent Magnet Synchronous Motors (PMSMs) are selected because they operate with a high degree of efficiency, are highly compact, high density of torque and better dynamic response. Conventional PMSM drive Mainstream PMSM drive systems would have the requirement of mechanical sensors, like encoders or resolvers, to obtain measurements of the rotor position and speed in order to employ the strategy of vector control. Use of such sensors, however, costs the system more, decreases reliability and adds complexity to the hardware especially in harsh industrial environments where sensors can fail because of vibration, temperature or electromagnetic interference. To overcome these shortcomings sensorless control methods have been developed as a useful alternative, such that the position and the speed of the rotor are determined by measuring the electrical quantities instead of the physical ones. This paper is an investigation on a sensorless approach to the control of PMSMs using observer-based adaptive estimation methods in order to have a correct estimation of rotor position and speed even without mechanical sensors. The approach proposed will combine a model-based state observer with an adaptive parameter estimation with an aim of raising the quality of estimation when the parameters vary, when subjected to load disturbances and changing operating environments. This control system is a hybrid of field-oriented control (FOC) and a nonlinear adaptive observer that allows accurate decoupling of the torque and flux components without compromising the quality of speed estimation at a large operating range. Simulation and experimental analysis of the implemented control strategy under various operating conditions are used to assess the performance of the proposed control strategy. These findings indicate that there is better estimation accuracy, quicker dynamic response, and smaller steady-state error than the traditional sensorless methods. All in all, the proposed approach is a valid, efficient, and economical solution of high-performance PMSM drive usage in industrial automation and electrical mobility systems.

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1. INTRODUCTION

1.1 Permanent Magnet Synchronous Motors Background

Permanent Magnet Synchronous Motors (PMSMs) are now seen as one of the most prevalent electrical machinery in the contemporary industrial drive programmes as they are highly efficient, have power density, are small in size and have excellent dynamic behaviour. Such benefits ensure the PMSMs are very applicable in sophisticated applications like electric vehicles, robotics, aerospace systems, wind energy

conversion systems, and computer numerical control (CNC) machines. When used in this way, high precision, efficiency and reliability require the proper control of the motor torque and speed. Permanent magnets eliminate rotor copper losses to achieve superior performance of the induction motors compared to conventional induction motors, as permanent magnets achieve better energy efficiency. Also with the ongoing pressure on performance motor drives in the industries, PMSMs have been developed as a popular solution to high-performance motion control.

1.2 Control of PMSM Drives as a Vector

In order to make the best use of the performance of PMSMs, some sophisticated methods of control are needed. Field-Oriented Control (FOC) also referred to as one of the most useful approaches in the contemporary operation of motor drives is the so-called vector control. The three-phase stator currents can be transformed into a rotating d-q reference frame by using vector control to provide independent control of motor torque and magnetic flux. This enables the PMSMs to be operated like independently excited DC motors, thus they respond dynamically quickly and control their torque exactly. In double rotor-based feedback systems such as those used in controlling vectors, rotor position and speed require precise information to synchronise the rotating reference frame correctly with the rotor magnetic field. Thus, the majority of the traditional type of vector control systems use rotor position sensors in order to receive real-time feedback data.

1.3 drawbacks of sensor-based control system

Rotor position and speed are typically detected by one of the following mechanisms with traditional PMSM control systems: mechanical There are several common types of mechanical sensors: encoders, resolvers, or Hall-sensors. Despite the fact that these sensors give good results, they also present a number of drawbacks to the motor drive system. Sensors add to a total cost of the system, area as well as the intricacy of the hardware and decrease the dependability of the system as a result of the potential sensor failures. Moreover, sensors need extra wiring and servicing, which might be inappropriate in severe conditions of industries with high temperatures, vibrations, or dust. Such restrictions have inspired researchers to devise other solutions that can satisfy the requirement of not using mechanical sensors but stay in control of high-performance motors.

1.4 Sensorless Control Observer-Based Estimation

In a bid to overcome the shortcomings of sensor-based control system sensorless control methods have more recently been actively studied. Sensorless correlates are sensorsless, forming an estimation of rotor position and speed based on electrical signals that are quantifiable (voltage and currents on the stator). Only some of the sensorless methods as observer-based estimation methods have received a

lot of attention as their robustness, high accuracy of the estimation, and capability to address the change in parameters as well as load perturbations. The adaptive estimation methods relying on observers are a combination of the system modelling and real-time parameter adjustment to enhance control behaviour in the varying working conditions. A sensorless approach of controlling PMSMs where an observer estimates rotor positions and speed is proposed in this study to improve the accuracy of the estimated speed and position. The inherent idea of the suggested approach is the combination of an adaptive observer and a framework of a vector control that ought to enable the attainment of stable and valuable motor drive performance under a broad spectrum of operating conditions.

2. LITERATURE REVIEW

Permanent Magnet Synchronous Motors (PMSMs) have found significant market interests in the current industrial drive systems because of their great efficiency, small-format-size, the high-power density, and dynamic behaviour. These properties allow using PMSMs in a large variety of applications including electric vehicles, robotics, renewable energy systems, and precision motion control. Traditional PMSM control methods use mechanical sensors like encoders and resolvers to get rotor position and speed data that is needed to drive the rotor under a vector control strategy. The associated costs however involved in using these sensors raise costs of the systems, decrease reliability and involve extra hardware complexity. Vibration, temperature and electromagnetic interference may also be sensitive to the sensors when working in rough industrial conditions. In turn, sensorless control techniques have become a popular subject of study to achieve high-performance motor control, but without mechanical sensors [1], [9].

The original sensorless control studies primarily centred on Back Electromotive Force (Back-EMF) based estimations. Those are the techniques of determining the position and speed of the rotor by the induction of a voltage in the windings of the stator as a result of the rotating magnetic field. The methods based on the back-EMF are comparably simple and work effectively in the middle-speed and high-speed of the motors at which the voltage induced is large enough. They however work poorly at low speeds as the magnitude of the back-EMF is very small and therefore the estimation accuracy is very

low and the operation is unstable. Hence the approaches cannot be used in those applications that need large range of speeds of operation [6].

In order to remove the drawbacks of the Back-EMF techniques, researchers proposed sensorless PMSM control methods that are based on Sliding Mode Observer (SMO) techniques. Sliding mode observers have excellent resilience to parameters uncertainties and external disturbances. They are also known to be fast converging and highly precise concerning estimation as opposed to the traditional modalities of estimation. Research has realised that SMO-based methods are able to predict rotor position effectively even when different loads are applied as well as when various parameters vary. Nevertheless, the key disadvantage of sliding mode observers is the appearance of chattering effects due to discontinuous switching functions that can introduce noise and decrease the accuracy of the estimation [11].

The Extended Kalman filter (EKF) is another sophisticated estimation technique in sensorless Permanent magnets machines drives that is widely researched. The EKF-based estimation is the estimation of the system states (rotor speed, position, and flux linkage) performed by using stochastic filtering methods. The technique gives the best estimation quality under the conditions of noise in measurement and modelling uncertainties. A number of studies have proved that sensorless control based on EKF can be operable within extensive working speed with high accuracy of estimation. Nevertheless, the major limitation to EKF techniques is that they are extremely computational intensive and thus, need a powerful digital processor to be realised in real time [7].

Besides the methods that use observers, the Model Reference Adaptive System (MRAS) approaches have also been widely analysed to estimate speed in motor drives. MRAS-based methods incorporate two of the models a reference model and an adaptive model to reconstruct rotor speed based on an adaptive process in minimising the difference between the two models. These are comparatively easy methods to apply, have sufficient estimation accuracy in steady-state operating conditions. Nevertheless, the techniques can be sensitive to changes in the parameters like the stator resistance caused by thermal effects [10], [12].

In the modern view, research has been done on sophisticated observer-based adaptive estimation methods in order to improve the performance of sensorless PMSM control systems further. Adaptive

observers also integrate the model-based estimation with adaptive adjustment of the parameters in order to enhance the ability to resist load disturbances, parameter non-certainty and non-linear dynamics in the system. The use of better nonlinear flux observer and extended-state observer has been suggested in order to increase stability in estimation and dynamic response of PMSM drives. These methods are more accurate as well as convergent than the traditional methods of estimators [8], [3].

Although technological advancements in sensorless PMSM control have seen some great improvements in sensorless control extending to the full range of speeds and the process of rotor position estimation requires minimal computation complexity, sensorless PMSM control is still difficult challenge. Thus, a key avenue of research in enhancing the performance, reliability and cost performance of the current PMSM drive systems, is the development of effective observer based adaptive estimation strategies.

3. METHODOLOGY

The research methodology offered targets the application of sensorless approach to the control of PMSMs that is based on the techniques of adaptive estimating the observer. This methodology involves three major steps and they include system modelling, adaptive observer design and control system implementation.

3.1 Mathematical PMSM Modelling.

Mathematical modelling of Permanent Magnet Synchronous Motor (PMSM) is needed in the analysis and design of advanced control strategies like sensorless vector control. Clarke transformations and Park transformations to make the control process simple the three phase stator quantities are converted to the two-axis rotating reference frame called the d-q reference frame. This conversion makes it possible to describe the motor variables in direct-axis (d-axis) and quadrature-axis (q-axis) components which turn at the same rate as the rotor magnetic field. This allows the multi-phase motor system with three phases to be reduced to two orthogonal units, with the opportunity to control the torque and the flux independently as in the case of a DC motor. The approach to modelling is very common in the contemporary PMSM control system, as it enhances control performance and simplifies the analysis.

The electrical dynamics of the PMSM in the d-q reference frame can be represented with stator voltage equations which are equations that make stator voltages, currents, and motor parameters dependent on one another. These equations use significant motor variables like stator resistance, d-axis and q-axis inductances, electrical angular speed and permanent magnet flux linkage. The relationship between stator voltage in the d q frame can be given in terms of d q as;

$$V_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q$$

where V_d represents the d-axis stator voltage, R_s denotes stator resistance, i_d and i_q are the d-axis and q-axis current components, L_d and L_q represent the stator inductances in the respective axes, and ω_e is the electrical angular speed of the motor. These electrical equations describe how voltage and current interact within the motor windings under rotating magnetic field conditions.

The PMSM model comprises the electromagnetic generation of torque and mechanical dynamics of the rotor in addition to the electrical dynamics. The major effect on the electromagnetic torque is the interaction between the stator currents and the permanent magnet flux. The balance between the electromagnetic torque, load torque and frictional losses determines the mechanical movement of the rotor. Such parameters as rotor inertia and damping coefficient define the rate at which the motor responds to changes in load or input voltage Figure 1. Combining the electrical and mechanical equations together, one can be confident that the PMSM is fully represented mathematically and can be used to design observer-based estimation algorithms and execute high-performance sensorless vector control schemes.

3.2 Adaptive Estimation of Observers

The observer-based methods of estimation are also important in sensorless control of Permanent magnet Synchronous Motors (PMSMs), estimating rotor position and speed without any mechanical sensors. In this method, the observer utilises measurable electrical quantities like stator voltages and currents in determining values about internal states of the motor that cannot be directly determined. The observer works based on the comparison of the recorded system outputs and the predicted outputs as the mathematical model of the motor to produce. Any discrepancy between the actual and estimated outputs is assumed to be an error in the estimation

that is utilised to correct the observer states. This will allow the observer to keep continuous track of the real motor states and give correct estimates which are necessary in the control algorithm.

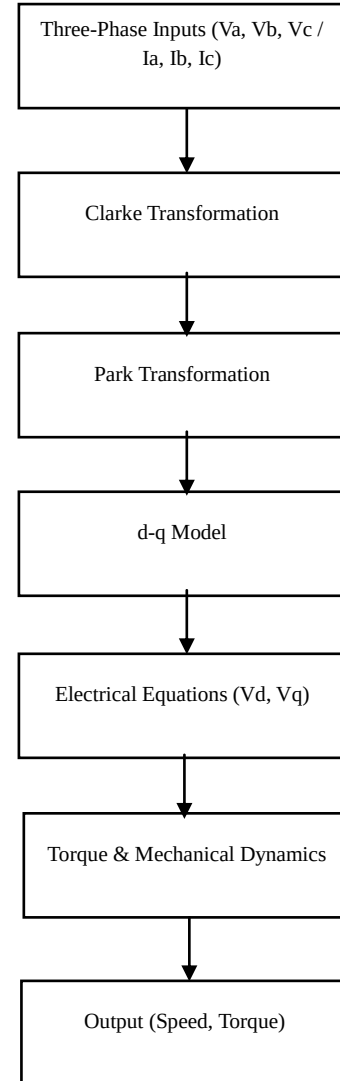


Fig. 1: Block Diagram of PMSM Mathematical Modeling Using Clarke and Park Transformations in the d-q Reference Frame

The mathematical modelling of the observer is founded on a state-space model of the PMSM dynamic behaviour. The motor model together with the feedback correction terms is used to update the estimated states by the observer. The equation of general observer can be written as:

$$\hat{\mathbf{x}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u + \mathbf{K}(y - \hat{y})$$

where $\hat{\mathbf{x}}$ represents the estimated state vector of the system, u denotes the system input, y represents the measured output, and $\hat{y}$ is the estimated output obtained from the observer model. The matrix $\mathbf{K}$ represents the observer gain that determines how

strongly the estimation error influences the correction of the estimated states. By properly selecting the observer gain, the estimation error can be minimized, allowing the observer to accurately track the actual motor states.

An adaptive mechanism is added to the observer design to estimate real time rotor speed to further enhance the performance of the estimation. The adjusted estimation procedure is constant and through the error signals caused by the disparity between the measured and the estimated variables

the speed is estimated. This adaptive scheme enables the observer to make up the variation in parameters, disturbances in loads and inaccuracy and imprecision in the model which might arise during the operation of the motor Figure 2. Consequently, the observer-based adaptive estimation technique improves the accuracy, robustness, and stability of sensorless PMSM control system, providing a generalised motor control at a significantly large operating range and operating conditions.

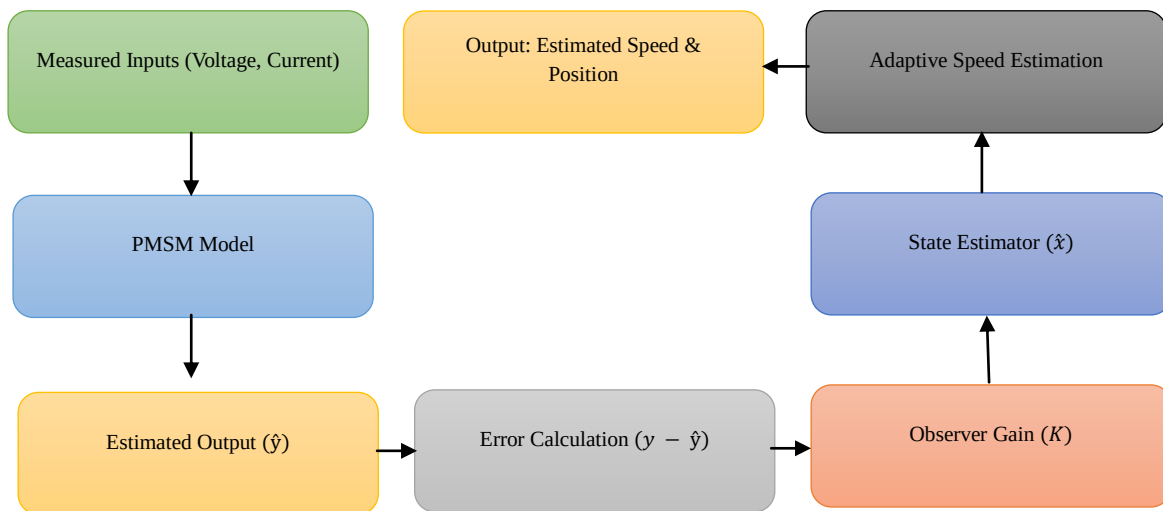


Fig. 2: Block Diagram of Observer-Based Adaptive Estimation for Sensorless PMSM Control

3.3 Implementation of Sensorless Vector Control

Adaptive Observer Refinement: Adaptive Observer and Vector Control

The sensorless control of vectors is effected by incorporating the adaptive observer and a Field-Oriented Control (FOC) system to give the Permanent Magnet Synchronous Motor (PMSM) in exact and efficient control. The things use a lot of control over torque and magnetic flux in the form of transforming the three-phase stator currents into a rotating dq reference frame, by the Clarke and Park transformations. By transforming the variables of AC motors into DC-like terms, this transformation makes the process of control quite easy since the motor torque and speed can be accurately regulated. Under the proposed system, stator voltages and current measured stator voltages and currents by the adaptive observer are used to estimate the rotor position as well as the rotor speed. Such estimated values take the place of the readings that would be

provided by mechanical sensors and the motor drive can be run without the encoders or resolvers and still provides good control behaviour.

Control Algorithms and system operation.

A sensorless type of the vector control system is made up of a few major components such as current control loops, a speed controller, an adaptive observer and a Pulse Width Modulation (PWM) inverter. The control process takes the stator current and voltages of the motor drive system that could be measured. The measured currents are then converted to the d-q reference frame by Clarke and Park transformations. These electrical signals are utilised by the adaptive observer to approximate the position and speed of the rotor in real time. According to the approximated rotor data, the speed controller produces reference current waveforms which control motor torque and flux. The current control loops read these reference signals and generate the desired voltage commands Figure 3. The last stage is the PWM inverter which divides the voltage commands into switching signals that move the motor. Based on this combined control approach, the PMSM is capable of

stabilised operation, enhanced dynamic response as well as high efficiency with no resort to mechanical sensors.

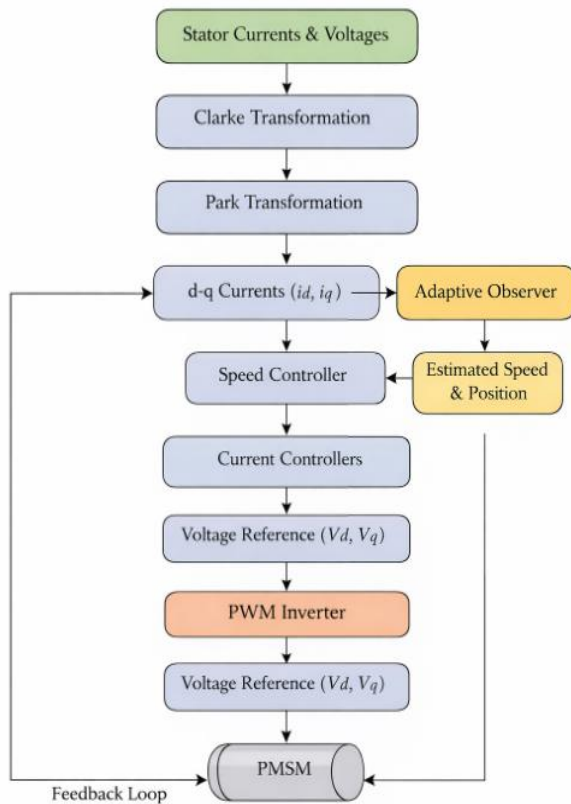


Fig. 3: Block Diagram of Sensorless Vector Control of PMSM Using Adaptive Observer and Field-Oriented Control (FOC)

4. RESULTS AND DISCUSSION

4.1 Simulation Environment and System Description

To conduct an evaluation of an effectiveness of the suggested sensorless control strategy, simulation experiments were carried out in the MATLAB/Simulink environment. Permanent Magnet Synchronous Motor (PMSM) drive system Permanent Magnet Synchronous Motor drive was considered based on realistic electrical and mechanical parameters such that the system would have performed as in real world operating conditions. Companies were required to include the following important elements in the simulation model PMSM mathematical model, inverter, vector control block, current regulators, speed controller, and the suggested observer-based adaptive estimation algorithm. Different working scenarios including starting up, variations of loads and steady-state performance were tested to

determine the robustness and precision with the proposed control method.

4.2 Rotor Speed and Position Estimation Performance

The results of the simulation show that the adaptive observer can estimate the speed and the position of the rotor accurately without mechanical sensors. In the startup condition the rapid convergence of the observer to the actual rotor speed is observed which implies quick dynamic response and effective estimation ability. The approximated speed is very close to the motor actual speed with very little variation and this proves the efficiency of the observer based estimation method. Such a fast convergence enhances the stability of a system and makes it controllable in reliable motor control even in the circumstances of transient operation.

4.3 Disturbance during Performance Under Load

The proposed control system was also tested regarding the disturbance in sudden loads. In case the motor externally received load torque, the control system was able to stabilise the functioning. The adaptive observer swiftly utilised the approximated rotor speed and compensated the disturbance so that the motor did not take off much time to resume its operating speed with minimum oscillation Figure 4. Also, there were no falls in torque responsiveness, which showed that the motor torque is properly controlled with a vector control strategy even when the load is varied.

4.4 Performance and Control Effectiveness Steady-State

The variation between the estimated rotor speed and American speed at steady-state operation was small implying that they exhibited high estimation accuracy. The findings also show the proposed adaptive estimation method using the observer is more robust to variations in parameters like the stator resistance which occurred due to changes in temperature Table 1. Additionally, the method of control of the vectors has been effective in decoupling torque and flux components, which allows one to control torque with great precision and provides the motor with a steady operation. In general, the simulation outcomes confirm the usefulness of the recommended sensorless vector control frameworks of PMSM drives, which makes it relevant to high-performance applications in the motor control of industrial motors.

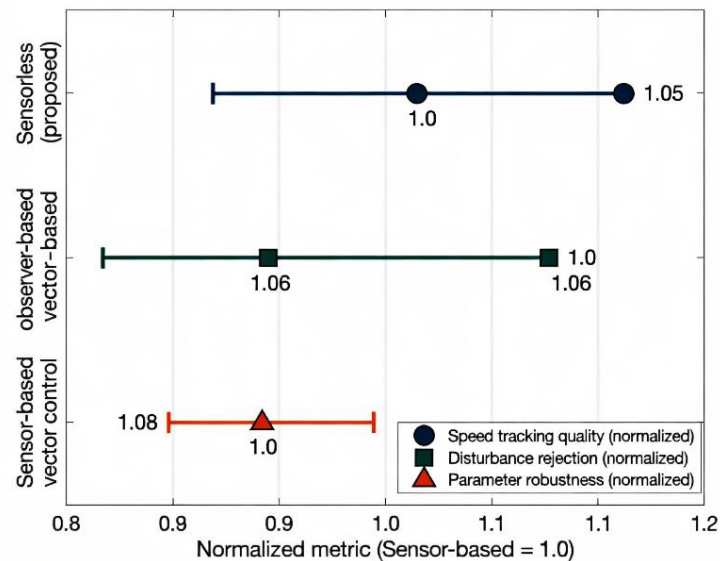


Fig. 4: Comparative Performance Analysis of Sensorless and Sensor-Based PMSM Control Methods Using Normalized Metrics

Table 1: Performance Evaluation of Proposed Sensorless PMSM Control System

Performance Parameter	Test Condition	Observed Result	Remarks
Rotor Speed Estimation Accuracy	Startup	Very high accuracy, minimal error	Fast convergence of adaptive observer
Dynamic Response	Startup	Rapid response	Quick tracking of reference speed
Speed Tracking Performance	Transient Condition	Closely follows actual speed	Negligible deviation observed
Load Disturbance Rejection	Sudden Load Change	Stable operation	Fast compensation by adaptive observer
Torque Response	Load Variation	Smooth and stable	No oscillations observed
Settling Time	Load Disturbance	Short settling time	Improved dynamic stability
Steady-State Error	Steady-State Operation	Very low	High estimation precision
Parameter Variation Robustness	Temperature Variation	High robustness	Resistant to stator resistance changes
Torque and Flux Decoupling	Normal Operation	Successfully achieved	Effective vector control performance
Overall System Stability	All Conditions	Stable	Reliable sensorless control operation

5. CONCLUSION

This paper has introduced a sensorless vector control approach of the Permanent Magnet Synchronous Motors (PMSMS) with an observer-based adaptive estimation method in order to avoid mechanical position sensors. To examine electrical and mechanical dynamics of the motor, a mathematical model of the PMSM was formulated in the $d-q$ rotating reference frame. In this model, an adaptive observer has been developed to determine the rotor position and speed by using available measurable electrical signals like stator voltages and stator currents. The viewer became involved in a Field-

Oriented Control (FOC) scheme in order to have precise torque and flux control and a stabilised functioning of the motor. The computer simulation results, which were conducted in MATLAB/Simulink, showed that the proposed approach has good estimations, high dynamic response, and high load disturbances and parameter variations resilience. The adaptive estimation system is good at minimising steady-state error and improving the stability of the overall control system. Thus, the suggested sensorless control of the vectors can be regarded as an effective, efficient, and cost-effective solution to the application of the PMSM drive to be used in industrial automation, electric vehicles, and robots. The further

work will be directed at the experimental validation of the study through real-time hardware implementation and the further optimization of the adaptive observer in the case of embedded motor drive companies.

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